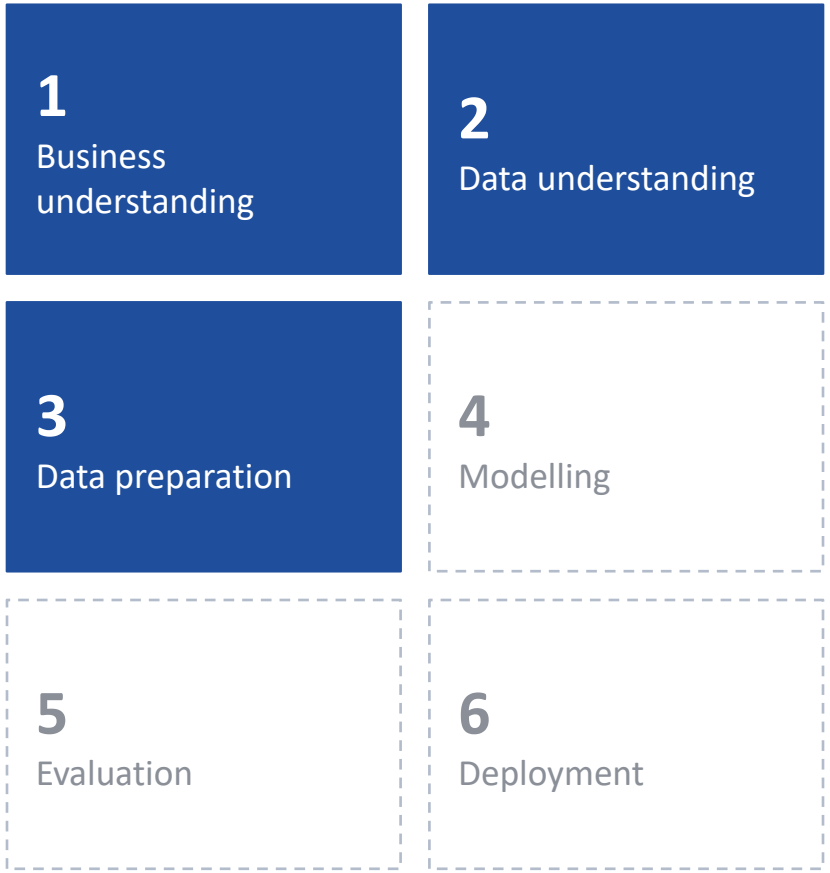
NOVA IMS
INFORMATION MANAGEMENT SCHOOL

Dashboard
https://github.com/MohamedTrikiSoftwareEngineer/DM_Group_10/tree/main

Data Mining Project · Deliverable 1

What separates ACDC's coworking members?

Exploratory data analysis, organised along CRISP-DM



Filled tiles = covered in this deliverable

Group 10
Gustavo Carvalho · Mohamed Triki · Diogo Gaspar

NOVA IMS
INFORMATION MANAGEMENT SCHOOL

ACDC wants to understand who its members are

Notebook section 1

The business problem

- Some members want their own desk and extra services. Others just want a cheap desk for a few days a week
- ACDC wants to know these groups, so it can adapt its services and marketing and keep members for longer

What we want from the data

- Nothing in the data says which group a member belongs to, so we have to find the groups ourselves
- We will look at how much each member is worth and how they use the space, then combine both

How we will know it worked

- For ACDC: we can describe each group in one sentence, and ACDC can treat that group differently
- For the data: the groups say more than the plan alone, and they stay similar if we clean the data a bit differently

15,465

rows and 31 columns in the raw file, one row per member

14,840

members left after cleaning

1 Jan 2024

the date we treat as "today". The file doesn't give one, so we worked it out from the data

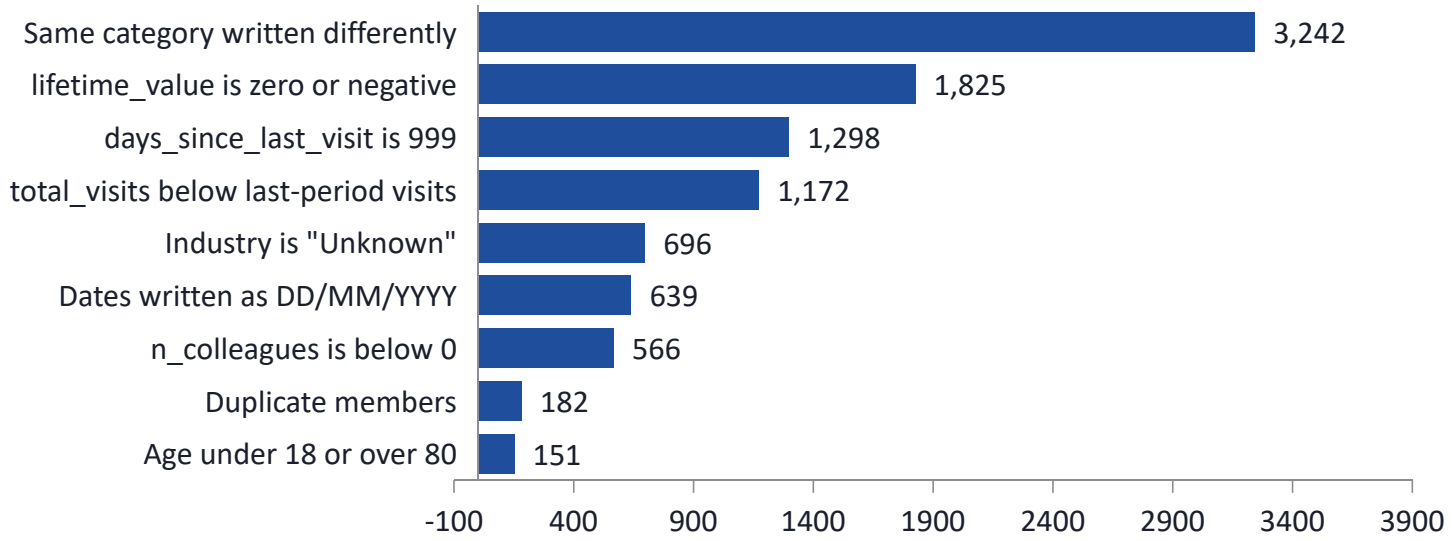
6

new columns we built from the data

The raw file was messy, so we cleaned it first

Notebook sections 2 and 3

Rows with each problem in the raw file (one row can have several)



71.75% of rows have no empty cell at all

25.6% have the end date 1970-01-01. We read this as "no end date yet", so these members count as active

18.2% have no usable lifetime_value after cleaning, up from 6.7%, because most zeros turned out to be hidden gaps

15,465 rows at the start

182 duplicates removed →

15,283 without duplicates

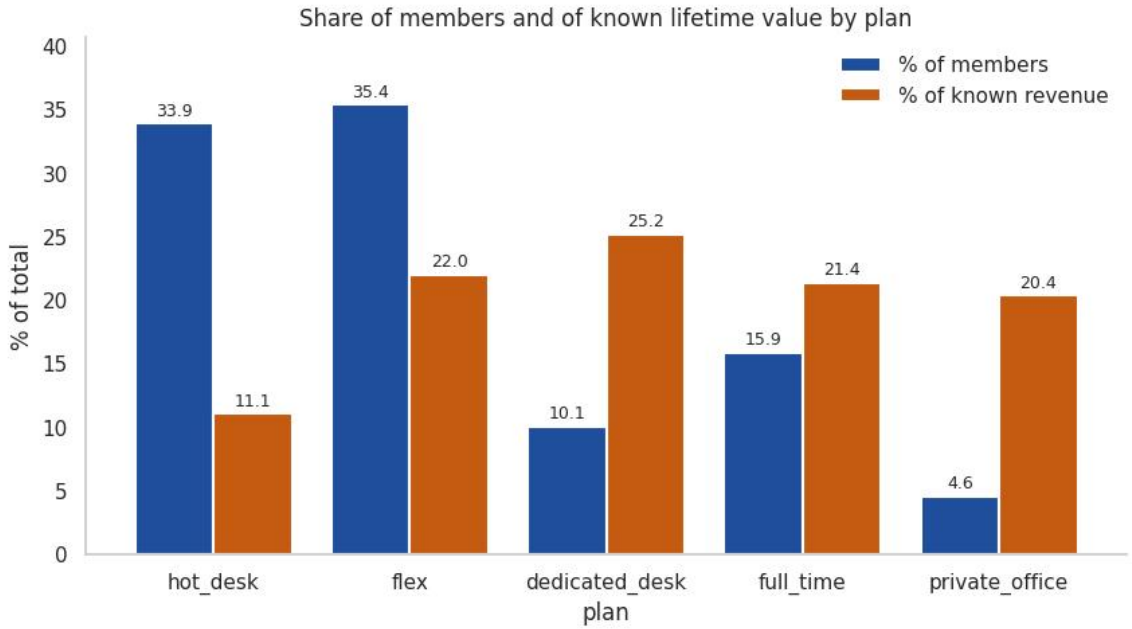
49 with no way to tell the plan →

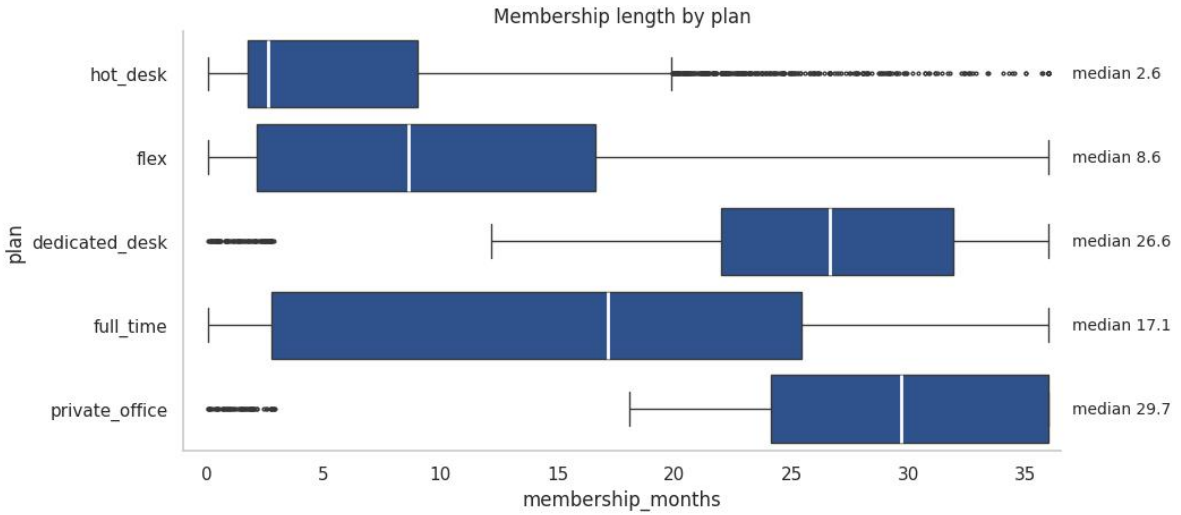
15,234 with a known plan

394 with dates that don't fit →

14,840 clean members

Most members do not bring most of the value Notebook sections 4.3 and 4.4





What we saw

Hot desk and flex have 69.3% of members but only 33.1% of the known value. A typical dedicated desk or private office member stays 26.6 and 29.7 months.

What it means

These two plans (dedicated desk and private office) bring more per member because they cost more and people stay longer. By our estimate, plans cost from 120 to 650 EUR a month.

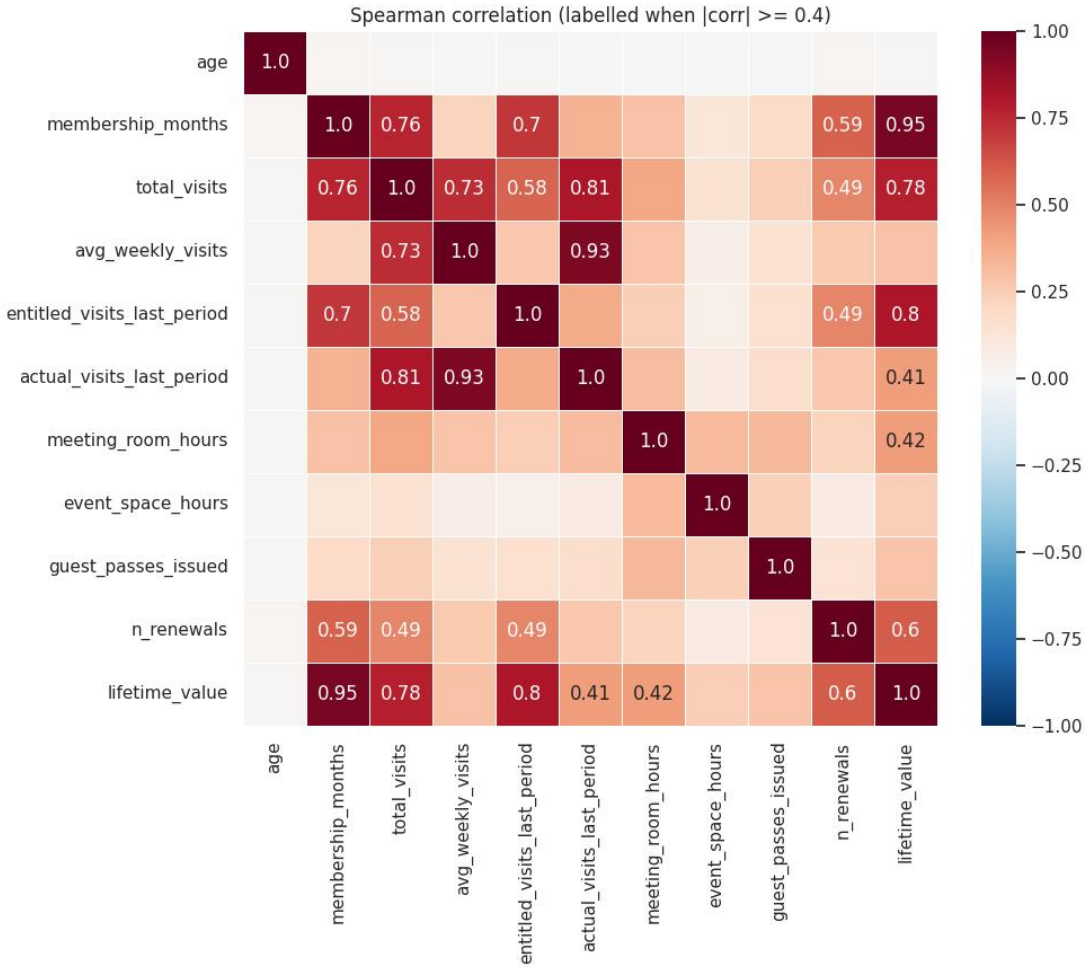
Why it matters to ACDC

ACDC could invest more in these plans. We only see revenue, though: private offices may cost more to build and take up more space.

What it means for the groups

We expect members to split into about 3 value groups, based on what they pay: cheaper, middle and top plans.

Members who stay longer visit more and bring more value Notebook section 4.9



0.95
Months as a member and lifetime value

0.78
Total visits and lifetime value

0.76
Months as a member and total visits

How to read this: 1 means two columns rise together. 0 means no link. No pair in this chart is negative, so no column falls while another rises.

What we saw

The longer someone is a member, the more visits they add up and the more they pay in total. Visits and value also rise together. Age has no real link with any of them.

What it means

Long memberships simply have more time to add up visits and money. So part of the link between visits and value is really just time.

Why it matters to ACDC

Value builds up month after month, so keeping members for longer should bring in more money.

What it means for the groups

If we use all these columns, the groups may only show how long people stayed. We will keep just some of them.

We expect about 3 value and 4 or 5 behaviour groups Notebook sections 5 and 6

Looking at	Groups we expect to find (still to be tested)
Value: about 3	Based on what members pay: hot desk and flex; full time and dedicated desk; private office
Behaviour: 4 or 5	New members; people who left early; people who left later; active members who still come; active members who have almost stopped (60.1% on dedicated desks)
Services	Mail handling and storage depend most on the plan (Cramér's V of 0.46 and 0.40), so they add less on their own

New column	What it tells us	Overlaps with
membership_months	How long they've been a member	total_visits (0.76)
plan_usage_rate	Share of allowed visits they used	avg_weekly_visits (0.84)
number_of_addons_used	How many extras they use	lifetime_value (0.53)
monthly_value	Money per month	lifetime_value (0.74)
days_since_last_visit_active	Days since an active member came	plan_usage_rate (−0.79)
plan_tier_ordinal	The plan as a number, 1 to 5	monthly_value (0.80)

What could go wrong

- The groups might just copy the plans, or only split new and old members
- Some values are still missing or wrong, and we must deal with them first
- Some columns say almost the same thing

What we will do next

- Build the value groups and the behaviour groups separately, then combine them
- Treat "uses a service" and "how much" as two different questions
- Fix the errors, then put very large values on a similar scale